4. VERIFICATION Of Thevenin's Theorem

AIM:-

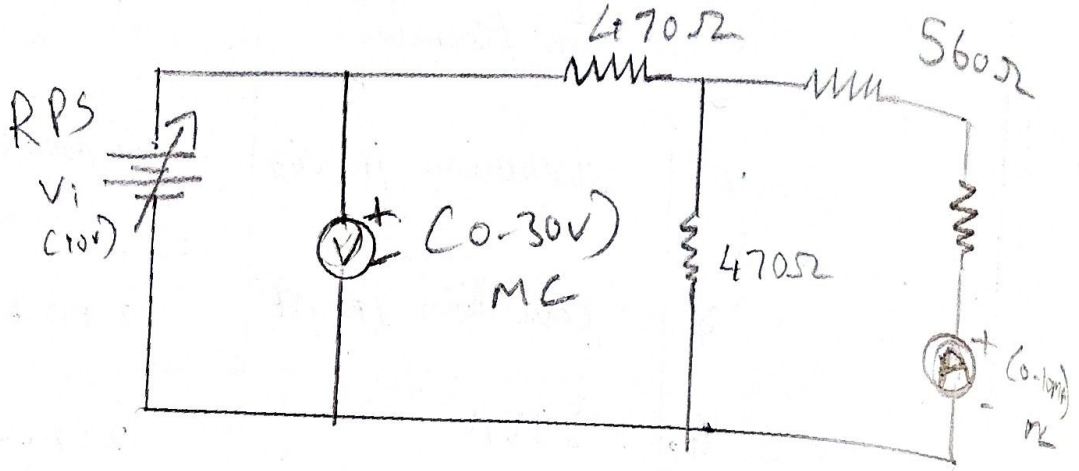
To verify Theorem Partically and Theoretically For the given DC circuit

APParatus Required:-

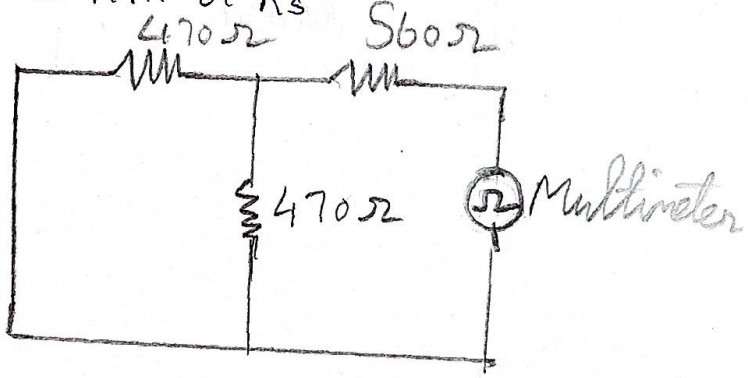
S.NO	APParatus	Specification	Quantity
1	Regulated Power supply (RPS)	(0-30)V	1
2.	Vollmeter	(0-30)VMC	1
3-	Ammeter	(0-10)MA MC	1
4.	Resistor	490Ω 570Ω 1kΩ	2 1 1
5.	Bread board	---	1
6	Multimeter	---	1

Circuit Diagram :- Verification of Thevenin Theorem

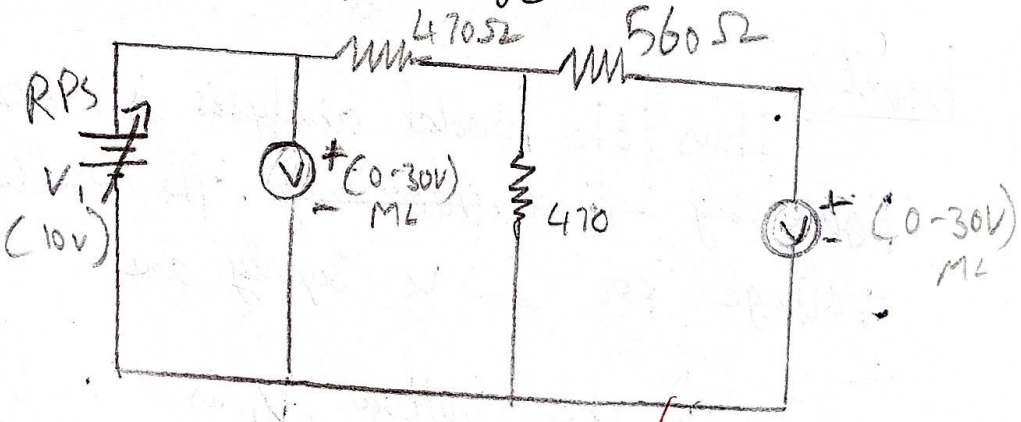
To Measure I_2



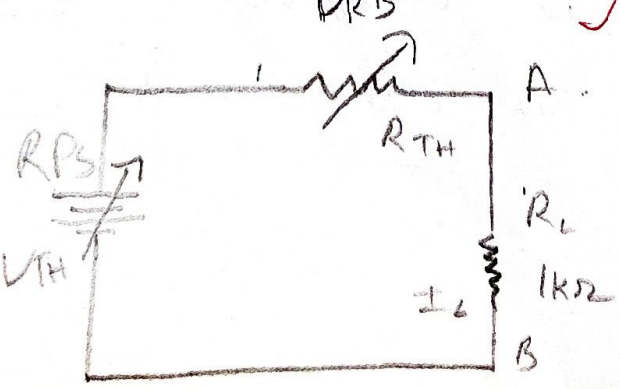
To measure R_{TH} or R_s



To measure V_{TH} or V_{OC}



Thevenin equivalent



Theory :-

THEVENIN'S THEOREM:-

Thevenin theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source, V_{TH} ($= V_{oc}$) in series with a resistor R_{TH}, where V_{TH} is the input or equivalent resistance at the load terminals (AB) when the independence sources are turned off.

Thevenin's voltage law:

It is open-circuit voltage across the load terminals (AB). i.e, $V_{TH} = V_{oc}$

Therenin's Resistance, R_{TH} :-

It is the look-back resistance across the load terminals when all the sources are replaced by short-circuit as its internal resistance is zero. An ideal current source is replaced by open-circuit as it internal resistance is infinity.

TABULAR COLUMN: 1

To Measure I_2

V_1 (Volt)	V_2 (AMPS)
10V	2.8MA

TABULAR COLUMN :- 2

To measure R_{TH} or R_N

R_{TH} (Ω)	R_N
795 Ω	785 Ω

TABULAR COLUMN :- 3

To Measure V_{TH} or V_{OC}

V_1 (volts)	V_{TH} (volts)
10V	5V

Procedure

1. Make the connections as per the circuit Diagram.

2. Vary the RPS and set an input voltage of 10 V

3. Note down the voltage reading (V_i) and ammeter reading (I_2) in tabular column.

4. Switch off the supply and make connection for circuit Diagram 2.

5. Measure the Thevenin's resistance R_{TH} = Norton's resistance R_N.

6. Switch off the supply and make connections for circuit Diagram: 3.

7. Set an input voltage of 10 V is the RPS and note down the voltmeter readings V_1 an V_{th} = (V_{oc}) in Tabular Column : 5

8. SWITH OFF the supply and make connections for circuit Diagram 4.

9. Set an input voltage of 10 V is the RPS and note down the voltage V_1 and Ammeter reading I_N (= I_{sc}) in Tabular Column 4.

Calculations :-

To find V_{TH}

By KCL

$$\frac{V_{TH}-10}{470} + \frac{V_{TH}}{470} = 0$$

$$2V_{TH} - 10 = 0$$

$$V_{TH} = 5V$$

By KVL,

$$10 = 470I + 470I$$

$$10 = 940\,I$$

$$I = 0.010A$$

$$V_{TH} = IR$$

$$= 0.010 \times 470$$

$$V_{TH} = 4.98$$

$$V_{TH} = 5V$$

To find R_{TH}.

$$R_{TH} = \frac{470 \times 490}{470 + 470} + 560$$

$$R_{TH} = 705\,\Omega$$

10) Draw the thevenin's equivalent circuit and Norton's equivalent circuit and nortons equivalent.

11) Calculate the I_L value using the formula thevenin's theorem.

$$I_L = V_{TH}/(R_{th} R_L)$$

Norton's theorem:

$$I_L = I_N * R_N (R_N + R_L)$$

12) Theoretically verify the norton's theorem

Marks of

Sl.No	Description	mark allotted	
1.	Experimental setup / connection	20 marks	20
2.	observation / Reading	20 marks	18
3.	calculating / Result	20 marks	18
4.	VIVA	20 marks	18
5.	Result	20 marks	18.
			92

Result:-

Thus Thevenins theorem is verified Practically and Theoretically.